

THEORY OF LINEAR MODELS

STAT 9120 | Fall 2026

Meeting	Tuesday & Thursday, 2:30-3:45 PM	Room	Health Sciences Building 1222
Instructor	Dustin Pluta, PhD	Email	dpluta@augusta.edu
Office	EB 212	Office Hours	By appointment
Credits	3	Prerequisite	STAT 7630 Applied Linear Models

Primary Text: Ravishanker, N., Chi, Z., and Dey, D. K. A First Course in Linear Model Theory, 2nd Edition. Chapman & Hall/CRC.

Catalog Course Description

This course is a study of the general linear statistical model. Topics include the theory of linear models, distributions of quadratic forms, full rank linear models, estimation under linear constraints and hypothesis testing for the full rank linear models, and fixed effect models of less than full rank, and other related models.

Course Objectives

1. Use matrix and vector-space methods to formulate and analyze general linear models.
2. Apply properties of symmetric, idempotent, projection, and generalized inverse matrices in linear model theory.
3. Derive and characterize the distributions of linear and quadratic forms of normal random variables.
4. Derive least-squares estimators and establish their statistical properties under the general linear model.
5. Distinguish between full-rank and less-than-full-rank linear models and explain the consequences of rank deficiency.
6. Determine whether linear functions of model parameters are estimable and construct estimators of estimable functions.
7. Obtain least-squares estimators subject to linear constraints and understand the relationship between constraints, identifiability, and estimability.
8. Formulate and test general linear hypotheses using quadratic forms and associated F-tests.
9. Construct and interpret confidence intervals and confidence regions for linear functions of model parameters.
10. Apply general linear model theory to fixed-effects models and related model formulations.

Course Evaluation and Grading

Assessments

Assessment	Weight
6 Homework Assignments	45%
Midterm Exam	25%
Final Exam	27%
Professionalism	3%

Homework Assignments

Homework assignments will emphasize the theoretical development of linear models and will consist primarily of mathematical derivations, proofs, and applications of matrix and distribution theory. Some assignments may also include computational problems intended to illustrate theoretical results.

Due dates will typically be 1-2 weeks after an assignment is given. Late assignments will not be accepted without prior approval.

Submitted assignments should be clearly organized, with the assignment number, student name, and date clearly visible on the first page. All derivations and supporting arguments should be presented clearly, including important intermediate steps rather than only a final result.

Any results, derivations, or other work included that is not the student's own must be cited with a complete reference. Typed assignments using LaTeX or Markdown are strongly preferred. Homework submission will occur online; see the course webpage for further instructions.

Midterm Exam

The midterm examination will be administered in class. The examination will emphasize mathematical and statistical reasoning and will include questions similar in style and difficulty to homework assignments and examples developed in lecture. Specific material covered and further details regarding the examination format will be provided in class.

Final Exam

The final examination will be cumulative, with an emphasis on material covered after the midterm. Students will be expected to demonstrate an integrated understanding of the theory of general linear models, including matrix methods, estimation, distributions of quadratic forms, estimability, and hypothesis testing. Specific details regarding the examination format will be provided in class.

Professionalism

For full credit in this category, students must consistently attend and participate in class and complete the course evaluations at the end of the term.

Grading Scale

Grade	Course Average
A	$\geq 90\%$ and $\leq 100\%$
B	$\geq 80\%$ and $< 90\%$
C	$\geq 70\%$ and $< 80\%$
D	$\geq 60\%$ and $< 70\%$
F	$< 60\%$

Course Topics & Schedule

Tentative. The schedule may be adjusted as needed; substantive changes will be communicated in class and through the course learning-management system.

Week / Dates	Topic	Coverage	Reading
Week 1 August 18 & 20	Review of Vector and Matrix Algebra	Vector spaces, subspaces, and linear independence; Rank, column spaces, and null spaces; Orthogonality and projections; Eigenvalues and eigenvectors; Matrix decompositions	Chapter 1
Week 2 August 25 & 27	Special Matrices and Quadratic Forms	Partitioned matrices; Symmetric and idempotent matrices; Positive definite and positive semidefinite matrices; Quadratic forms; Projection matrices and geometric interpretation	Chapter 2
Week 3 September 1 & 3	Generalized Inverses and Linear Systems	Generalized inverses; Solutions of consistent linear systems; Nonunique solutions; Rank-deficient systems; Connections with least-squares estimation	Chapter 3

Week / Dates	Topic	Coverage	Reading
Week 4 September 8 & 10	The General Linear Model	Formulation of the general linear model; Mean and error structure; Least-squares estimation and normal equations; Geometry of least squares; Projection onto the column space of the design matrix <i>No class September 8 - university follows Monday class schedule.</i>	Chapter 4
Week 5 September 15 & 17	Normal Theory for Linear Models	Multivariate normal distribution; Linear transformations of normal random variables; Independence of normal linear forms; Sampling distributions arising in linear models	Chapters 5-6
Week 6 September 22 & 24	Distributions of Quadratic Forms	Expectations and variances of quadratic forms; Chi-square distributions of quadratic forms; Independence of quadratic forms; Independence of linear and quadratic forms; Cochran's theorem and related results	Chapters 5-6
Week 7 September 29 & October 1	Estimation in Full-Rank Linear Models	Least-squares estimation; Unbiasedness; Variance and covariance of estimators; Gauss-Markov theorem and best linear unbiased estimation; Maximum likelihood estimation under normality	Chapter 7
Week 8 October 6 & 8	Statistical Inference for Full-Rank Models	Estimation of error variance; General linear hypotheses; Extra sums of squares; F-tests; Confidence intervals and confidence regions	Chapters 7-8
Week 9 October 13 & 15	Midterm Examination and Further Inference	Midterm examination; Power and noncentral distributions; Simultaneous inference; Multiple comparisons	Chapter 8
Week 10 October 20 & 22	Multiple Linear Regression as a General Linear Model	Regression within the general linear-model framework; Partitioned regression; Partial regression; Nested models; Tests of subsets of regression coefficients	Chapter 9
Week 11 October 27 & 29	Less-Than-Full-Rank Linear Models	Rank-deficient design matrices; Nonuniqueness of parameter estimates; Estimable functions and characterization of estimability; Generalized-inverse solutions	Chapter 10
Week 12 November 3 & 5	Linear Constraints and Identifiability	Estimation subject to linear constraints; Side conditions; Reparameterization; Equivalence of parameterizations; Relationship among constraints, identifiability, and estimability	Chapters 3, 4, and 10
Week 13 November 10 & 12	Hypothesis Testing in Rank-Deficient Models	Testable hypotheses; General linear hypotheses in less-than-full-rank models; Sums of squares; Construction and interpretation of F-tests; Invariance with respect to parameterization	Chapter 10

Week / Dates	Topic	Coverage	Reading
Week 14 November 17 & 19	Fixed-Effects Models	One-way fixed-effects models; Multi-factor fixed-effects models; Main effects and interactions; Contrasts; Analysis of variance as a general linear model; Balanced and unbalanced designs	Chapter 10
Week 15 November 24 & 26	Fixed-Effects Models, Continued	Additional fixed-effects models; Estimability in factorial models; Hypothesis construction in unbalanced designs; Connections between ANOVA and general linear model theory <i>No class November 26 - Thanksgiving Holiday.</i>	Chapter 10
Week 16 December 1 & 3	Related Models and Comprehensive Review	Introduction to random- and mixed-effects models; Generalized least squares and nonstandard covariance structures; Connections to extensions of the general linear model; Comprehensive review <i>Last class meeting: December 3.</i>	Selected material from Chapter 11 and instructor notes

Policies & Guidelines

Students are required to follow and be aware of the Augusta University Student Attendance Policy. Regular and punctual attendance is expected in all courses, and attendance and ongoing participation may be monitored throughout the semester.

Students are responsible for all university policies governing academic honesty, accommodations, student conduct, attendance, and other academic matters. Current Augusta University policies apply to this course.

Course materials, assignments, announcements, and any modifications to the tentative schedule will be communicated through the course learning-management system and/or during class.

The instructor reserves the right to make reasonable modifications to the course schedule, assignment due dates, or coverage of individual topics as necessary. Any such changes will be communicated to students in a timely manner.